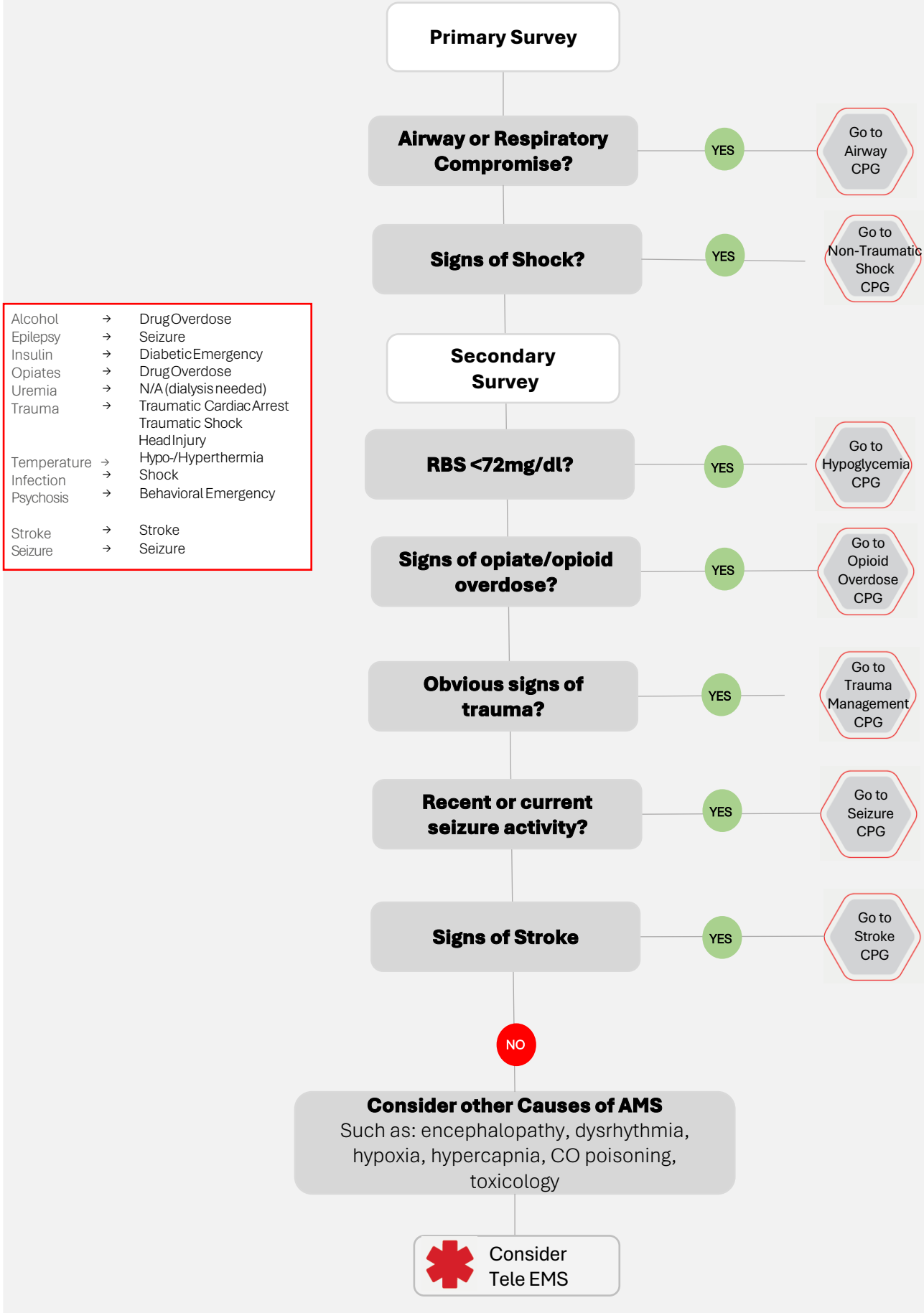
Altered Mental Status



Patient Care Goals

1. Identify and manage life-threatening causes of altered mental status.
2. Stabilize the patient's condition through appropriate interventions to restore normal mental status or prevent further deterioration.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Patients presenting with altered mental status, including:
  - Confusion, disorientation, or impaired responsiveness.
  - Lethargy, agitation, or unresponsiveness.
  - Inability to follow commands or respond appropriately to stimuli.
  - Symptoms ranging from mild confusion to coma.
- Suspected underlying causes such as hypoglycemia, hypoxia, stroke, overdose, infection, or trauma.

Exclusion Criteria

- None noted.

Key Patient Assessment Points

1. **Neurological Status Assessment**
   - Evaluate level of consciousness using AVPU (Alert, Verbal, Painful, Unresponsive) or Glasgow Coma Scale (GCS).
   - Assess for focal neurological deficits (e.g., unequal pupils, motor weakness, or speech impairment).
   - Note behavioral changes (e.g., agitation, combativeness, or lethargy).
2. **History**
   - Identify potential causes through SAMPLE history:
     - Recent events (e.g., trauma, drug use, or medication changes).
     - Medical history (e.g., diabetes, epilepsy, stroke, or psychiatric disorders).
     - Medication use (e.g., insulin, sedatives, or missed doses).
   - Check for medical alert bracelets or documentation indicating conditions like diabetes or epilepsy.
3. **Differential Diagnosis**
   - Assess for possible causes using the AEIOD TIPS pneumonic:
     - A – Alcohol, Abuse, Atypical migraine
     - E – Epilepsy, Electrolytes
     - I – Insulin (hypoglycemia)
     - O – Oxygen, Overdose
     - U – Uremia (kidney failure)
     - T – Trauma, Tumor
     - I – Infection
     - P – Psych, Poisoning
     - S – Seizure, Subarachnoid hemorrhage, Sepsis
4. **Glucose and Oxygen Assessment**
   - Check random blood sugar (RBS) to identify hypoglycemia (<70 mg/dL) or hyperglycemia (>200 mg/dL).
   - Assess for hypoxia via pulse oximetry (SpO2 <94%).
5. **Environmental and Toxicological Clues**
   - Look for evidence of drug or alcohol use (e.g., pill bottles, drug paraphernalia).
   - Assess for environmental exposures (e.g., carbon monoxide, chemicals).
6. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.
   - Red Priority:
     - GCS < or equal to 9, or progressive deterioration in GCS
     - Unresponsive or responsive only to pain

Patient Management & Interventions

1. **Primary and Secondary Survey**
   - Perform primary and secondary survey as per Universal Care CPG.
2. **Consider Red Flags**
   - Unresponsiveness or coma (AVPU).
   - Hypoglycemia (RBS <70 mg/dL) go to Hypoglycemia CPG.
   - Severe hyperglycemia (RBS >400 mg/dL) go to Hyperglycemia CPG.
   - Hypoxia (SpO2 <90%) or respiratory distress.
   - Suspected stroke (e.g., focal neurological deficits, sudden onset) go to Stroke CPG.
   - Signs of overdose (e.g., pinpoint pupils, respiratory depression) go to Overdose CPG.
3. **Consider ALS back-up**
   - Consider the requirement for ALS back-up as per Universal Care CPG.
4. **Evaluate for Signs of Shock or Instability**
   - Hypotension (SBP <90 mmHg in adults, or age-specific for pediatrics), tachycardia, or poor perfusion (e.g., cool, clammy skin, delayed capillary refill >2 seconds).
   - If shock is present, initiate fluid resuscitation and address underlying causes.
   - Manage Shock as per appropriate Shock Guideline.
5. **Establish IV Access**
   - Indicated for hypotension, suspected hypoglycemia, or anticipated medication administration.
   - Consider intraosseous (IO) access if IV access cannot be obtained
6. **Patient Reassessment**
   - Frequently Reassess the patient as per Universal Care CPG.
   - Specific Patient Reassessment Points:
     - Neurological Status
       - Reassess AVPU or GCS frequently to monitor for deterioration or improvement.

Patient Safety Considerations

- It is difficult to accurately assess patients with an altered mental status. All patients with an altered mental status should be transported to hospital.

Clinical Insights & Other Guidance

- None noted

Documentation

- Identification: Full patient name, age, and gender.
- Mechanism / Medical Complaint: Nature and progression of altered mental status, suspected underlying cause (e.g., hypoglycemia, overdose, stroke).
- Injuries / Information: Clues to contributing causes (e.g., trauma, alcohol, drug paraphernalia), presence of alert bracelets.
- Signs: GCS or AVPU scale, pupil size/reactivity, vital signs (BP, HR, SpO_2, temperature), RBS results.
- Treatment:
  - Glucose administration: route, dose, and timing.
  - Oxygen therapy and SpO_2 levels.
  - IV fluids or medications if given.
  - Response to treatment and changes in mental status.
- Allergies: Any known allergies, especially medications.
- Medications: Recent or chronic medications (e.g., insulin, sedatives, psychiatric drugs).
- Background: Known conditions such as diabetes, epilepsy, stroke, psychiatric illness.
- Other:
  - Environmental or toxicological clues.
  - Scene information and bystander reports.

Key Performance Indicators

1. Perform and document an RBS on all patients with altered mental status.
2. Assess and document the initial AVPU or GCS score.

References

1. Ambulance Victoria (2024) *Clinical practice guidelines*. Ambulance Victoria. Available at: https://www.ambulance.vic.gov.au/clinical-practice-guidelines.
2. Health Professions Council of South Africa (2018) *Clinical practice guidelines*. Health Professions Council of South Africa. Available at: https://www.up.ac.za/media/shared/62/ZP_Files/clinical_practice_guidelines_protocols_2018.zp160355.pdf.
3. Joint Royal Colleges Ambulance Liaison Committee (2019) *JRCALC clinical practice guidelines 2019*. Class Professional Publishing.
4. Kupas, D.O., Wydro, G.C., Tan, D.K., Kamin, R. and Harrell, A.J. (2018) Altered mental status: current evidence-based recommendations for prehospital care. *Prehospital Emergency Care*, 22(3), pp. 299-306.
5. National Association of State EMS Officials (2022) *National model EMS clinical guidelines*. National Association of State EMS Officials. Available at: https://flemsc.emergency.med.jax.ufl.edu/wordpress/files/2023/08/National-Model-EMS-Clinical-Guidelines_2022.pdf.
6. Ontario Ministry of Health (2023) *Basic life support patient care standards*. Ontario Ministry of Health. Available at: https://files.ontario.ca/moh_2/moh-standards-basic-life-support-patient-care-standards-v3-4-en-2023-03-10.pdf.
7. St John New Zealand (2023) *Clinical procedures and guidelines - comprehensive edition*. St John New Zealand. Available at: https://www.stjohn.org.nz/globalassets/documents/health-practitioners/clinical-procedures-and-guidelines---comprehensive-edition.pdf.